HW8: AWS Infrastructure with Packer & Terraform

Custom AMI (Amazon Linux 2023 + Docker)

VPC + Bastion Host + 6 Private EC2 Instances

Manmohan Sharma

March 2026

Architecture: Internet -> Bastion (public subnet, your IP only)
-> 6 Private Instances (private subnet, bastion access only)
NAT Gateway for outbound internet from private subnet

Region: us-west-2 | Instance Type: t3.micro
Repo: github.com/manmohan659/hw8

1. Terraform Init (Starting)

```
manmohan@Mac terraform % terraform init
Initializing the backend...
Initializing modules...
- bastion in modules/bastion
- ec2_private in modules/ec2-private
- routing in modules/routing
- security_groups in modules/security-groups
- subnets in modules/subnets
- vpc in modules/vpc
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
```

2. Terraform Init (Complete)

```
manmohan@Mac terraform % terraform init
Initializing the backend...
Initializing modules...
- bastion in modules/bastion
- ec2_private in modules/ec2-private
- routing in modules/routing
- security_groups in modules/security-groups
- subnets in modules/subnets
- vpc in modules/vpc
Initializing provider plugins...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
manmohan@Mac terraform %
```

3. Terraform Apply - Plan (Key Pair + Bastion)

```
Terraform will perform the following actions:

  # aws_key_pair.deployer will be created
  + resource "aws_key_pair" "deployer" {
      + arn             = (known after apply)
      + fingerprint     = (known after apply)
      + id              = (known after apply)
      + key_name        = "hw8-deployer-key"
      + key_name_prefix = (known after apply)
      + key_pair_id     = (known after apply)
      + key_type        = (known after apply)
      + public_key      = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPzbeqDRFCar0cKeMCz7pp+Fk
      + tags_all        = (known after apply)
    }

  # module.bastion.aws_instance.bastion will be created
  + resource "aws_instance" "bastion" {
      + ami                                  = "ami-014d82945a82dfba3"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
```

4. Terraform Apply - Plan (Instance Details)

```
Terraform will perform the following actions:

  # aws_key_pair.deployer will be created
  + resource "aws_key_pair" "deployer" {
      + arn             = (known after apply)
      + fingerprint     = (known after apply)
      + id              = (known after apply)
      + key_name        = "hw8-deployer-key"
      + key_name_prefix = (known after apply)
      + key_pair_id     = (known after apply)
      + key_type        = (known after apply)
      + public_key      = "ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPzbeqDRFCar0cKeMCz7pp+FknFT+dmTv1NiB3JJMUTO manmohan659@gmail.com"
      + tags_all        = (known after apply)
    }

  # module.bastion.aws_instance.bastion will be created
  + resource "aws_instance" "bastion" {
      + ami                                  = "ami-014d82945a82dfba3"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + cpu_core_count                       = (known after apply)
      + cpu_threads_per_core                 = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t3.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "hw8-deployer-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
      + placement_partition_number           = (known after apply)
      + primary_network_interface_id         = (known after apply)
      + private_dns                          = (known after apply)
      + private_ip                           = (known after apply)
```

5. Terraform Apply - Plan Summary (24 Resources)

```
  }

# module.vpc.aws_vpc.main will be created
+ resource "aws_vpc" "main" {
    + arn                                  = (known after apply)
    + cidr_block                           = "10.0.0.0/16"
    + default_network_acl_id               = (known after apply)
    + default_route_table_id               = (known after apply)
    + default_security_group_id            = (known after apply)
    + dhcp_options_id                      = (known after apply)
    + enable_dns_hostnames                 = true
    + enable_dns_support                   = true
    + enable_network_address_usage_metrics = (known after apply)
    + id                                   = (known after apply)
    + instance_tenancy                     = "default"
    + ipv6_association_id                  = (known after apply)
    + ipv6_cidr_block                      = (known after apply)
    + ipv6_cidr_block_network_border_group = (known after apply)
    + main_route_table_id                  = (known after apply)
    + owner_id                             = (known after apply)
    + tags                                 = {
        + "Name" = "devops-final-vpc"
      }
    + tags_all                             = {
        + "Name" = "devops-final-vpc"
      }
  }

Plan: 24 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + bastion_public_ip    = (known after apply)
  + private_instance_ips = [
      + (known after apply),
      + (known after apply),
      + (known after apply),
      + (known after apply),
      + (known after apply),
      + (known after apply),
    ]
  + vpc_id               = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
```

6. Terraform Apply - Resources Creating

```
aws_key_pair.deployer: Creation complete after 1s [id=hw8-deployer-key]
module.routing.aws_eip.nat: Creation complete after 2s [id=eipalloc-0c452cf9646b37c6a]
module.vpc.aws_vpc.main: Still creating... [00m10s elapsed]
module.vpc.aws_vpc.main: Creation complete after 13s [id=vpc-0c540fface21be213]
module.vpc.aws_internet_gateway.igw: Creating...
module.subnets.aws_subnet.public[1]: Creating...
module.subnets.aws_subnet.private[1]: Creating...
module.subnets.aws_subnet.public[0]: Creating...
module.subnets.aws_subnet.private[0]: Creating...
module.security_groups.aws_security_group.bastion: Creating...
module.vpc.aws_internet_gateway.igw: Creation complete after 1s [id=igw-05dc963b725ac91ce]
module.routing.aws_route_table.public: Creating...
module.subnets.aws_subnet.private[0]: Creation complete after 1s [id=subnet-0818c7f8940c8e022]
module.subnets.aws_subnet.private[1]: Creation complete after 1s [id=subnet-0746c5b04ba14e306]
module.routing.aws_route_table.public: Creation complete after 1s [id=rtb-03026d12a89495c54]
module.security_groups.aws_security_group.bastion: Creation complete after 3s [id=sg-07ca1923f6955bcc6]
module.security_groups.aws_security_group.private: Creating...
module.security_groups.aws_security_group.private: Creation complete after 3s [id=sg-0a3738bdb3ce363c8]
module.ec2_private.aws_instance.private[1]: Creating...
module.ec2_private.aws_instance.private[3]: Creating...
module.ec2_private.aws_instance.private[0]: Creating...
module.ec2_private.aws_instance.private[2]: Creating...
module.ec2_private.aws_instance.private[4]: Creating...
module.ec2_private.aws_instance.private[5]: Creating...
module.subnets.aws_subnet.public[1]: Still creating... [00m10s elapsed]
module.subnets.aws_subnet.public[0]: Still creating... [00m10s elapsed]
module.subnets.aws_subnet.public[0]: Creation complete after 11s [id=subnet-0a6233c26d51a4c81]
module.subnets.aws_subnet.public[1]: Creation complete after 11s [id=subnet-06a355af038acb3d4]
module.routing.aws_route_table_association.public[1]: Creating...
module.routing.aws_route_table_association.public[0]: Creating...
module.routing.aws_nat_gateway.nat: Creating...
module.bastion.aws_instance.bastion: Creating...
module.routing.aws_route_table_association.public[1]: Creation complete after 1s [id=rtbassoc-0bf5d74f5e608ea1c]
module.routing.aws_route_table_association.public[0]: Creation complete after 1s [id=rtbassoc-0276da545734b57c5]
module.ec2_private.aws_instance.private[1]: Still creating... [00m10s elapsed]
module.ec2_private.aws_instance.private[3]: Still creating... [00m10s elapsed]
module.ec2_private.aws_instance.private[0]: Still creating... [00m10s elapsed]
module.ec2_private.aws_instance.private[2]: Still creating... [00m10s elapsed]
module.ec2_private.aws_instance.private[4]: Still creating... [00m10s elapsed]
module.ec2_private.aws_instance.private[5]: Still creating... [00m10s elapsed]
```

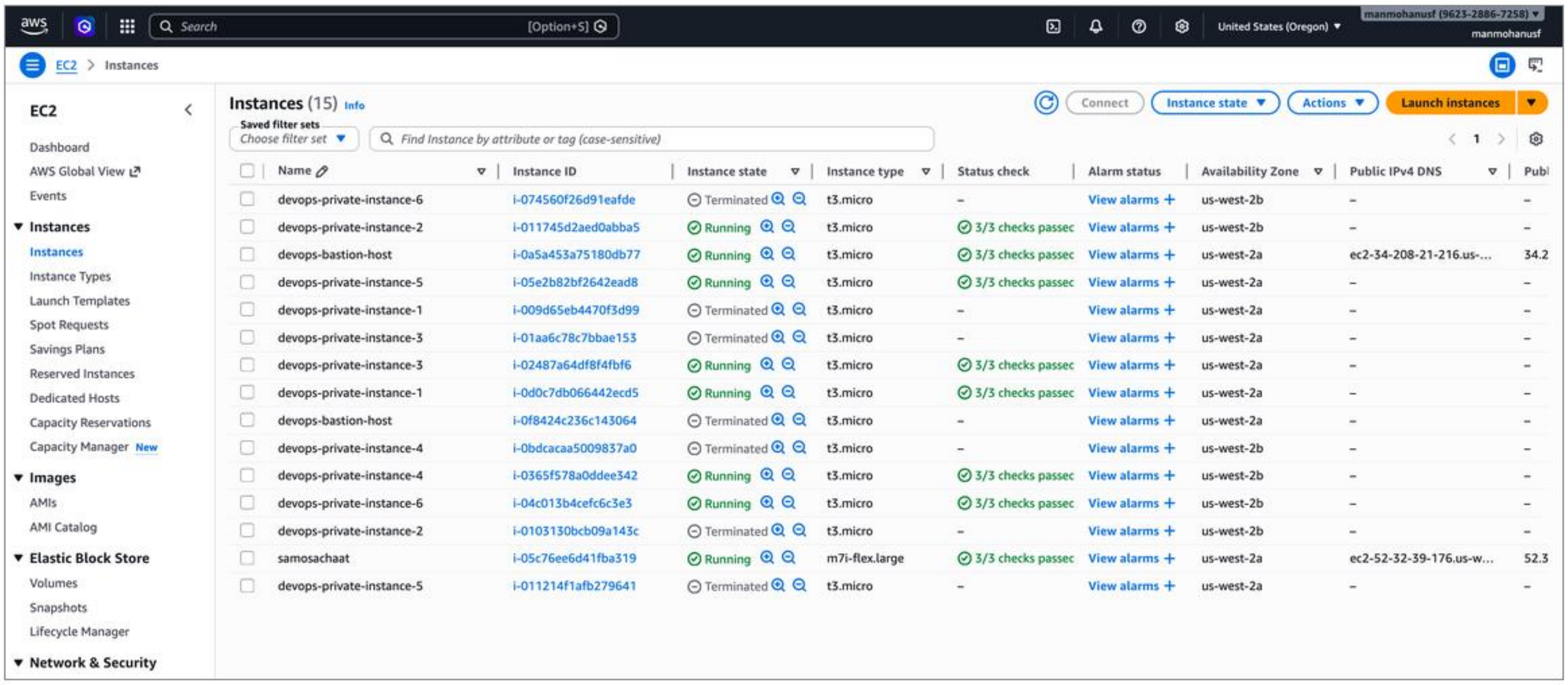
7. Terraform Apply Complete - All 24 Resources Created

```
Apply complete! Resources: 24 added, 0 changed, 0 destroyed.

Outputs:

bastion_public_ip = "34.208.21.216"
private_instance_ips = [
  "10.0.10.46",
  "10.0.11.83",
  "10.0.10.142",
  "10.0.11.73",
  "10.0.10.182",
  "10.0.11.78",
]
vpc_id = "vpc-0c540fface21be213"
manmohan@Mac terraform %
```

8. AWS Console - EC2 Instances (7 Running)



9. SSH Bastion Hop to Private Instance - Docker Verified

```
manmohan@Mac terraform % ssh -A -i ~/.ssh/id_ed25519_github ec2-user@34.208.21.216
The authenticity of host '34.208.21.216 (34.208.21.216)' can't be established.
ED25519 key fingerprint is: SHA256:cJja7JnSwE/or1uEWl0ZI5h3dB4HoSoBpKBDvqPOW9c
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.208.21.216' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '→
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-0-1-144 ~]$ ssh ec2-user@10.0.10.46 'echo "Connected to private instance!" && docker --version && uname -a'
The authenticity of host '10.0.10.46 (10.0.10.46)' can't be established.
ED25519 key fingerprint is SHA256:BSG6dDgmbDcd3R6gU6S4FxyChufq4XytSAE2a6E3EGU.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.10.46' (ED25519) to the list of known hosts.
Connected to private instance!
Docker version 25.0.14, build 0bab007
Linux ip-10-0-10-46.us-west-2.compute.internal 6.1.164-196.303.amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Fri Mar  6 16:11:23 UTC 2026 x86_64
x86_64 x86_64 GNU/Linux
[ec2-user@ip-10-0-1-144 ~]$
```

10. Terraform Destroy - Cleanup

```
Last login: Sun Mar 29 21:18:26 on ttys008
manmohan@Manmohans-MacBook-Air ~ % cd ~/Documents/FinalSemester/DevOps/hw8/terraform
manmohan@Manmohans-MacBook-Air terraform % terraform destroy -auto-approve
module.subnets.data.aws_availability_zones.available: Reading...
module.bastion.data.aws_ami.amazon_linux: Reading...
aws_key_pair.deployer: Refreshing state... [id=hw8-deployer-key]
module.routing.aws_eip.nat: Refreshing state... [id=eipalloc-0c452cf9646b37c6a]
module.vpc.aws_vpc.main: Refreshing state... [id=vpc-0c540fface21be213]
module.subnets.data.aws_availability_zones.available: Read complete after 0s [id=us-west-2]
module.bastion.data.aws_ami.amazon_linux: Read complete after 1s [id=ami-014d82945a82dfba3]
module.vpc.aws_internet_gateway.igw: Refreshing state... [id=igw-05dc963b725ac91ce]
module.subnets.aws_subnet.private[0]: Refreshing state... [id=subnet-0818c7f8940c8e022]
module.subnets.aws_subnet.private[1]: Refreshing state... [id=subnet-0746c5b04ba14e306]
module.subnets.aws_subnet.public[1]: Refreshing state... [id=subnet-06a355af038acb3d4]
module.subnets.aws_subnet.public[0]: Refreshing state... [id=subnet-0a6233c26d51a4c81]
module.security_groups.aws_security_group.bastion: Refreshing state... [id=sg-07ca1923f6955bcc6]
module.routing.aws_route_table.public: Refreshing state... [id=rtb-03026d12a89495c54]
module.routing.aws_nat_gateway.nat: Refreshing state... [id=nat-05aed2d0a36e2d538]
module.routing.aws_route_table_association.public[1]: Refreshing state... [id=rtbassoc-0bf5d74f5e608ea1c]
module.routing.aws_route_table_association.public[0]: Refreshing state... [id=rtbassoc-0276da545734b57c5]
module.security_groups.aws_security_group.private: Refreshing state... [id=sg-0a3738bdb3ce363c8]
module.bastion.aws_instance.bastion: Refreshing state... [id=i-0a5a453a75180db77]
module.routing.aws_route_table.private: Refreshing state... [id=rtb-02120645fab43b74e]
module.ec2_private.aws_instance.private[5]: Refreshing state... [id=i-04c013b4cefc6c3e3]
module.ec2_private.aws_instance.private[2]: Refreshing state... [id=i-02487a64df8f4fbf6]
module.ec2_private.aws_instance.private[0]: Refreshing state... [id=i-0d0c7db066442ecd5]
module.ec2_private.aws_instance.private[3]: Refreshing state... [id=i-0365f578a0ddee342]
module.ec2_private.aws_instance.private[1]: Refreshing state... [id=i-011745d2aed0abba5]
module.ec2_private.aws_instance.private[4]: Refreshing state... [id=i-05e2b82bf2642ead8]
module.routing.aws_route_table_association.private[1]: Refreshing state... [id=rtbassoc-010d531d750cc1536]
module.routing.aws_route_table_association.private[0]: Refreshing state... [id=rtbassoc-03340c34896984e19]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
symbols:
  - destroy

Terraform will perform the following actions:

  # aws_key_pair.deployer will be destroyed
```